AMBER
Apparatus for Meson and Baryon Experimental Research

Current Status of the Simulation of Radiative Effects with McMule

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McMule

Project Overview and Progress

- Topic: Simulation of Radiative Effects for the AMBER Proton Radius Measurement with McMule
- Learning to work with McMule with support from Marco Rocco
- Simulation of LO $(\mu+p\rightarrow\mu+p)$ and NLO processes
- Especially investigating the real-photon emission process
- Investigation of the impact of different cuts on the kinematic

- Cuts used to obtain the results in this presentation:
  - $0.3<\theta_\mu<2.[mrad]$
  - $E_\mu>70\,GeV$
  - $-12.<\theta_\gamma<12.[mrad]$

Expected Cross Sections and Count Rates

$$0.001 < Q^2 < 0.04\,[GeV^2/c^2]$$

Photon Energy	100 MeV	200 MeV	500 MeV
σ_{LO} [mb]	0.250566	0.250567	0.250562
R_{LO} [1/s]	31.90232	31.90239	31.90178
$\sigma_{R\gamma}$ [mb]	0.000115	0.000106	0.000090
$R_{R\gamma}$ [1/s]	0.014614	0.013475	0.011401
σ_{NLO} [mb]	0.002864	0.002855	0.002839
σ_{FULL} [mb]	0.253430	0.253422	0.253401

Count rate input values:

Target length: 60 cm

Hydrogen pressure: 20 bar

2.687 x10^{19} protons/cm^3

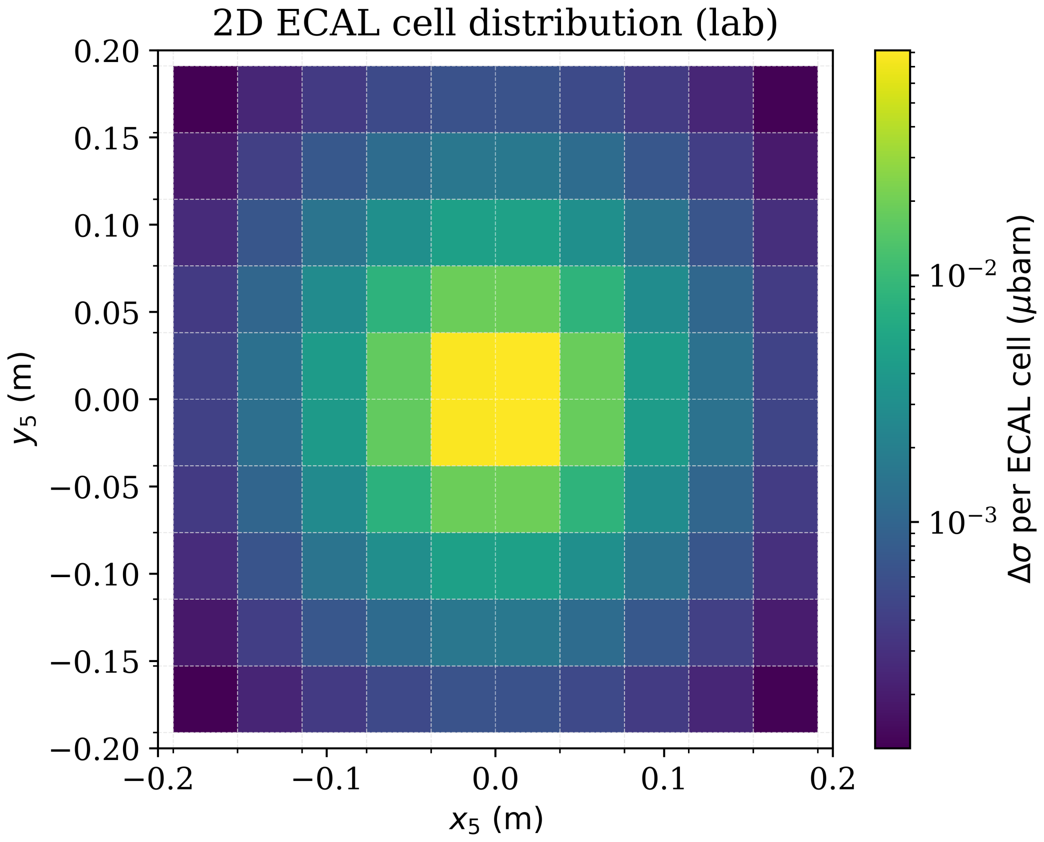
Beam intesity: 2 x10^6 1/s

$$0.3 < \theta_\mu < 2.\,[mrad]$$
$$E_\mu > 70\,GeV$$
$$-12. < \theta_\gamma < 12.\,[mrad]$$

- Increasing the photon-energy threshold reduces both the real-photon cross section and count rate
- The real-photon process contributes at the ~0.04% level (depending on photon energy threshold)

BACKUP

Expected Photon Cross Section



$0.3 < \theta_\mu < 2.[mrad]$
$E_\mu > 70\, GeV$
$-12. < \theta_\gamma < 12.[mrad]$

Used Equations

- ```
  def calculate_rate(sigma_mb):
    ltarget = 60 #cm
    Hpres = 20 #bar
    Npvol = 2.687*1e19 #Protons/cm^3
    Ibeam = 2*1e6 #1/s
    Np = Npvol * 2 * (Hpres/1.013) * ltarget      #Protons/cm^2 (target thickness)

    rate = sigma_mb * 1e-27 * Np * Ibeam
    return rate #1/s
  ```
  ```